Trial (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
1(a)	$6\sqrt{2}$	2	M1 for $\sqrt{6^2+6^2}$
1(b)	$3\sqrt{2}$	1	B1 for $\frac{1}{2}\times 6\sqrt{2}$
1(c)	45°	2	M1 for $\tan(\angle VPM)=\frac{3\sqrt{2}}{3\sqrt{2}}$
2(a)	$\frac{5x+8}{12}$	2	M1 for correctly finding a common denominator and expanding numerators $\frac{9x-4(x-2)}{12}$
2(b)	$\frac{x+3}{x+4}$	3	B1 for factoring numerator as $(x-3)(x+3)$
2			B1 for factoring denominator as $(x-3)(x+4)$
2			A1 for cancelling the common factor to achieve final answer
3(a)	$\frac{9}{4}$ or $2\frac{1}{4}$	2	M1 for converting to improper fraction $\left(\frac{27}{8}\right)^{\frac{2}{3}}$
3(b)(i)	$\frac{b}{2a^2}$ or $\frac{1}{2}a^{-2}b$	3	B1 for coefficient $\frac{1}{2}$, B1 for a^{-2}, B1 for b
3(b)(ii)	$2^m\times 2^3+2^m$	3	M1 for separating the index on 2^{m+3}
3	$2^m(2^3+1)$		M1 for correctly factoring out 2^m
3	$2^m(8+1)=9\times 2^m$		A1 for explicit valid sequence to the required format
4(a)	2^2 or 4	1	B1
4(b)	6^3 or 216	1	B1
4(c)	$2\frac{1}{4}$	2	M1 for $\left(\frac{3}{2}\right)^2$ or $\frac{1}{\left(\frac{2}{3}\right)^2}$ or $\frac{2^{-2}}{3^{-2}}$ or $\frac{9}{4}$
5	$x=\frac{5}{y-1}$	3	M1 for clearing denominator: $yx=x+5$
5			M1 for isolating and factorising: $x(y-1)=5$

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6(a)	$\frac{60}{360} \times 2 \times \pi \times 6 = 2\pi$	2	M1 for $\frac{60}{360} \times 2 \times \pi \times 6$
6(b)	30π	2	M1 for $\frac{360-60}{360} \times \pi \times 6^2$
7(a)	$\frac{d}{\frac{\left(\frac{d}{3}\right)}{x} + \frac{\left(\frac{2d}{3}\right)}{2x}} = \frac{3}{2}x$	3	M1 for total time $= \frac{d}{3x} + \frac{2d}{6x}$
7			M1 for simplifying denominator to $\frac{2d}{3x}$
7			A1 for $\frac{d}{\left(\frac{2d}{3x}\right)} = \frac{3x}{2}$
7(b)	48	2	M1 for equating $\frac{3}{2}x = \frac{270}{3\frac{3}{4}} = \frac{270 \times 4}{15} = 72$
8(a)	90	1	
8(b)	56	1	
8(c)	56	1	M1 for alternate angles to $\angle ACB$
8(d)	34	1	M1 for angles in the same segment
9(a)	$\frac{3}{5}$	1	B1 for correct gradient
9(b)	$y = \frac{3}{5}x + 4$	2	M1 for $y - (-2) = \frac{3}{5}(x - (-10))$ or $-2 = \frac{3}{5}(-10) + c$
9(c)	$(0, 4)$	1	B1 ft from their c value in part (b)
10(a)	Gradient of given line is $\frac{3}{4}$	M1	
10	Perpendicular gradient is $-\frac{4}{3}$	M1	
10	$3y = -4x - 7$ or $3y = -4x + c$ and substitutes $(2, -5)$	A1	Correct integer form
10(b)	$A\left(-\frac{7}{4}, 0\right)$ and $B\left(0, -\frac{7}{3}\right)$	B2	B1 for each correct point

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10(c)	Area = $\frac{1}{2} \times \left\lvert -\frac{7}{4} \right\rvert \times \left\lvert -\frac{7}{3} \right\rvert = \frac{49}{24}$	A2	M1 for $\frac{1}{2} \times$ their x-intercept $\times$ their y-intercept
11(a)	483.71	1	B1
11(b)	484	1	B1
12(a)	$x^3 + 4x^2 - 5x$	3	M2 for $x(x^2 + 4x - 5)$, M1 for $x^2 - x$ or $x^2 + 4x - 5$
12(b)	$(y-3)(y-5)$	2	M1 for $(y+a)(y+b)$ where $ab = 15$ or $a + b = -8$
12(c)	$7ab(2a-b)$	2	M1 for $7a(2ab - b^2)$ or $7b(2a^2 - ab)$ or $ab(14a - 7b)$
13(a)	2000 AUD	2	M1 for $16000 \times \left(\frac{1}{2}\right)^3$ or 8000 then 4000 seen
13(b)	14000 AUD	1	B1 for 16000 − ans from (a)
14(a)	$\frac{13}{16} = \frac{13 \times 625}{16 \times 625} = \frac{8125}{10000} = 0.8125$ or long division shown cleanly.	2	M1 for multiplying by a factor of $\frac{625}{625}$ or for demonstrating unambiguous long division A1 for complete proof of structural identity
14(b)	$A = B$, C, D or $\frac{13}{16} = 0.8125$, 81.3%, $\frac{41}{50}$	3	M1 for converting values to common numeric platform: $C = 0.8130$, $D = 0.8200$ M1 for at least two values correctly ranked relative to each other
15			
15(a)	Correct 3 boundary lines drawn	2	M1 for drawing at least 2 correct lines
15	Correct unwanted regions shaded	1	
15(b)	$(2,2)$, $(2,3)$, $(2,4)$, $(2,5)$, $(3,3)$, $(3,4)$, $(3,5)$, $(4,4)$, $(4,5)$, $(5,5)$	2	B1 for finding at least 5 correct points with no incorrect points

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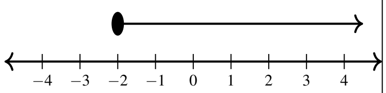
16*	45 cm^2	2	M1 for area ratio $= \left(\frac{6}{4}\right)^2$ or $\left(\frac{3}{2}\right)^2$ or $\frac{9}{4}$ seen or implied
17(a)	$(1,-1)$	2	M1 for $\left(\frac{-2+4}{2}, \frac{-5+3}{2}\right)$ or one coordinate correct
17(b)	Radius $= \frac{1}{2}\sqrt{(4-(-2))^2+(3-(-5))^2} = \frac{1}{2}\sqrt{6^2+8^2} = \frac{1}{2}(10) = 5$	2	M1 for finding diameter $= 10$ or using distance from centre to C or D
18(a)	$x \geq -2$	4	M1 for correctly identifying common denominator and multiplying to remove fractions e.g. $9(x-2)-4(2x+1) \geq -24$, M1 for correct expansion to $9x-18-8x-4 \geq -24$, M1 for properly isolating the variable $x-22 \geq -24$
18(b)	 **18**	2	B1 for solid circle strictly at -2, B1 for arrow pointing in the correct positive direction
18(c)	-2	1	B1 for -2 only
19(a)	350 000 mm^3	1	B1 for 350×1000
19(b)	6.755 kg	2	M1 for $350 \times 19.3 = 6755$ g and dividing by 1000
19(c)	2.8 cm^3	2	M1 for $350 \times \left(\frac{1}{5}\right)^3$ or $\frac{350}{125}$
20(a)(i)	$-\frac{3}{4}$	2	M1 for rearranging to $4y = -3x+12$ or $y = -\frac{3}{4}x+3$
20(a)(ii)	3 or $(0,3)$	1	B1 for correct intercept
20(b)	$x = 5$	1	B1 for correct vertical line equation
21(a)	$\frac{3}{8}$, 0.35, $\frac{7}{20}$, 32%	2	M1 for converting numbers to a common form: $0.375, 0.35, 0.32, 0.35$
21(b)	$\geq$ or $\leq$	1	B1 for recognizing both sides evaluate to 4, thus satisfying either inclusive boundary relation

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22(a)	2646 EUR	2	M1 for Year 1: $2400 \times 1.05 = 2520$
22(a)(i)			M1 for Year 2: their $2520 \times 1.05 = 2646$ with no dots on line
22(a)(ii)	132.30 EUR	2	M1 for $2778.30 - 2646$ or $2646 \times \frac{5}{100}$
23(a)	Correct proof to $x = 4$	3	M1 for $2(2x-1)+x+4$, M1 for $2(x+3+x)$, A1 for explicit simplification $5x+2 = 4x+6 \implies x = 4$
23(b)	28	2	M1 for evaluating dimensions $(4+3)\times 4$
24(a)	50	2	M1 for 25×2 or for identifying $5^2 = 25$ or $\sqrt[3]{8} = 2$
24(b)	988	2	M1 for $1000 - 12$ or for identifying $10^3 = 1000$ or $\sqrt{144} = 12$
25(a)	$n = 200$	3	M1 for actual area $= 12 \times 16 = 192\ \text{m}^2$, M1 for converting to cm^2: $192 \times 10000 = 1920000\ \text{cm}^2$, A1 for $n^2 = \frac{1920000}{48} = 40000 \implies n = 200$
25(b)	27 cm	2	M1 for $\frac{5400}{200}$ oe
26(a)	23	1	B1
26(b)	$4n+3$	2	B2 for correct answer, or B1 for $4n+k$ (where k is any constant or omitted)
27(a)	$\text{n}(A \text{ only}) = 40 - 4x$ $\text{n}(B \text{ only}) = 37 - 5x$ $\text{n}(C \text{ only}) = \text{n}(\text{none}) = 40 - 4x$ Sum: $(40-4x)+(37-5x)+(40-4x)+(2x-3)+(x+1)+2x+x+(40-4x) = 155-11x = 100$	3	M1 for expressing individual single regions in terms of x M1 for full summation expression equal to 100 A1 for flawless verification reaching $155 - 11x = 100$

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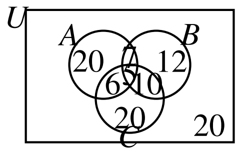
27(b)	 Left: 20, Right: 12, Bottom: 20, Top middle: 7, Bottom-left middle: 6, Bottom-right middle: 10, Centre: 5, Outside: 20	3	B1 for solving $x = 5$ and finding the four intersection regions B1 for correct single-sport regions (20, 12, 20) B1 for correct outer region (20)
28(a)	$x = \frac{5}{2}$ or $x = -\frac{5}{2}$	2	M1 for $(2x-5)(2x+5) = 0$ or $x^2 = \frac{25}{4}$
28(b)	$x = \frac{3}{2}$	2	M1 for $(2x-3)^2 = 0$
29(a)	7×10^2	3	M1 for numerator simplified to 14×10^4 or 1.4×10^5, M1 for division step $\frac{1.4 \times 10^5}{2 \times 10^2}$ or $\frac{140\,000}{200}$
30(a)	2	1	B1 for correct answer
30(b)	2	2	M1 for $\frac{3-(-3)}{2-(-1)}$ or equivalent
31(a)	$7\sqrt{3}$	2	B1 for $5\sqrt{3}$ or $2\sqrt{3}$
31(b)	$2\sqrt{5}$	2	M1 for $\frac{10\sqrt{5}}{5}$
32(a)	$9 - \frac{9}{4}\pi$	3	M1 for area of square $= 3 \times 3 = 9$
32			M1 for area of sector $= \frac{90}{360} \times \pi \times 3^2$
32(b)	960π	3	M1 for linear scale factor $= \frac{12}{3} = 4$
32			M1 for volume scale factor $= 4^3 = 64$
33(a)	-24	1	A1 for correct value
33(b)	$2\frac{3}{8}$ or $\frac{19}{8}$	2	M1 for $\frac{7}{4} + \frac{5}{8} = \frac{14}{8} + \frac{5}{8}$
34(a)	10:45 p.m.	1	B1 for correct time format with p.m.
34(b)(i)	19 20	2	M1 for adding 9 hours and 11 hours 35 minutes to 22 45 or equivalent

Part	Answer	Marks	Partial Marks / Guidance
34(b)(ii)	Wednesday	1	B1 for Wednesday
35(a)	$126 = 2 \times 3^2 \times 7$ and $540 = 2^2 \times 3^3 \times 5$	2	B1 for each correct prime factorisation
35(b)	$18x^2y^2$	2	M1 for finding $\mathrm{HCF}(126, 540) = 18$ or for selecting x^2y^2
35(c)	$3780x^4y^5z^3$	2	M1 for finding $\mathrm{LCM}(126, 540) = 3780$ or for selecting $x^4y^5z^3$